

Module 24.09: Regularization — Ridge Regression (L2 Norm)

1. Mathematical Formulation of Ridge Regression (Tikhonov Regularization)

Ridge Regression penalizes the squared Euclidean norm (L2 norm) of the model weights:

$$J_{\text{Ridge}}(\mathbf{w}, b) = \frac{1}{2m} \sum_{i=1}^m \left(\hat{y}^{(i)} - y^{(i)} \right)^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

Where:

$$\|\mathbf{w}\|_2^2 = \sum_{j=1}^p w_j^2 = \mathbf{w}^T \mathbf{w}$$

The factor of $\frac{1}{2}$ cancels out the exponent 2 during differentiation.

2. Analytical Closed-Form Solution: The Ridge Normal Equation

Unlike Lasso (whose L1 absolute value has non-differentiable cusps requiring numerical coordinate descent), Ridge Regression is continuously differentiable everywhere. Differentiating $J_{\text{Ridge}}(\mathbf{w})$ with respect to $\mathbf{w}$:

$$\nabla_{\mathbf{w}} J_{\text{Ridge}} = \frac{1}{m} \mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w} = 0$$

Solving for $\mathbf{w}$:

$$(\mathbf{X}^T \mathbf{X} + m\lambda \mathbf{I}) \mathbf{w} = \mathbf{X}^T \mathbf{y}$$

Standardizing the formulation (absorbing m into λ):

$$\mathbf{w}_{\text{Ridge}}^* = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

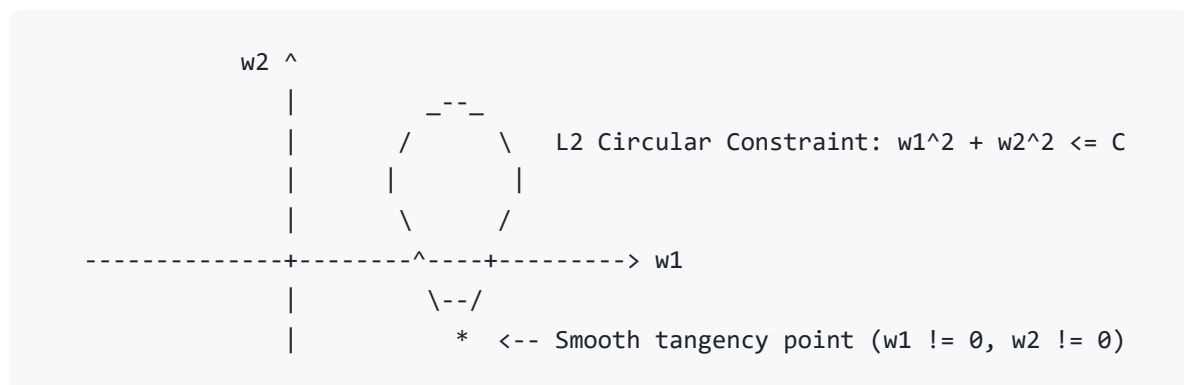
Mathematical Salvation from Multicollinearity:

In standard OLS, if features are multicollinear, $\mathbf{X}^T \mathbf{X}$ is singular and non-invertible. In Ridge Regression:

- We add $\lambda \mathbf{I}$ (where $\lambda > 0$) along the principal diagonal.
- All eigenvalues σ_i are shifted by $+\lambda$: $\sigma_i(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}) = \sigma_i + \lambda > 0$.
- The matrix is guaranteed to be strictly positive definite and invertible! Ridge always has a unique, stable analytical solution.

3. Geometric Intuition: Smooth Shrinkage vs. Sparsity

In parameter space, the constraint region $|\mathbf{w}|_2^2 \leq C$ forms a smooth circle (or hypersphere in higher dimensions):



Because the boundary of a circle is completely smooth with no sharp corners, the loss contour makes contact at an arbitrary point along the curve:

- Weights are shrunk smoothly toward zero ($w_j \rightarrow 0$), but are **almost never set strictly to zero**.
- Ridge retains all features, reducing their variance while preserving their informational contribution.